

11. What is the difference between = and ==?

Answer:

- = is the assignment operator. It assigns a value to a variable.
- == is the comparison operator. It checks whether two values are equal.

Example:

```
x = 10
```

```
print(x == 10) # True
```

12. What are Python operators?

Answer:

Operators are symbols used to perform operations on variables and values.

Types:

Arithmetic Operators

- +, -, *, /, %, //, **

Comparison (Relational) Operators

- ==, !=, >, <, >=, <=

Assignment Operators

- =, +=, -=, *=, /=, %=, //=, **=

Logical Operators

- and, or, not

Bitwise Operators

- &, |, ^, ~, <<, >>

Membership Operators

- in, not in

Identity Operators

- is, is not

13. What is an if statement?

Answer:

An if statement executes a block of code only if a condition is true.

Example:

```
age = 18  
  
if age >= 18:  
    print("Eligible to vote")
```

14. What is the difference between if, elif, and else?

Answer:

- if checks the first condition.
- elif checks another condition if the previous one is false.
- else executes when none of the conditions are true.

15. What is a nested if statement?

Answer:

A nested if is an if statement inside another if statement.